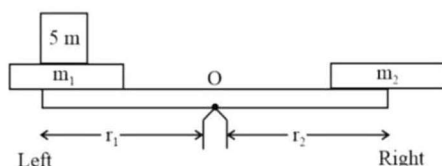


TOPIC : ROTATIONAL MOTION

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A.	2	1	2	1	1	4	1	3	1	1	1	3	1	2	1
Q.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	4	1	1	1	2	1	3	4	3	3	4	4	4	1
Q.	31	32	33	34	35	36	37	38	39	40					
A.	3	1	1	4	3	4	2	2	3	2					

HINT & SOLUTION

1. ANS - 2



Let m_1 and m_2 be the masses of pan

It is given that $m_1 = m_2$

Since balance is achieved so net torque about o should be zero

$$(5mg + m_1g) r_1 = (m_2g) r_2$$

$$(5mg + m_1g) r_1 = (m_1g) r_2$$

$$\frac{r_1}{r_2} = \frac{m_1g}{5mg + m_1g}$$

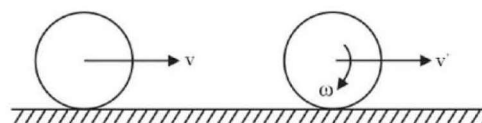
$$\frac{r_1}{r_2} < 1$$

2. ANS - 1

When a body rolls down without slipping along an inclined plane of inclination θ , it rotates about a horizontal axis through its centre of mass and also its centre of mass moves. Therefore, rolling motion may be regarded as

a rotational motion about an axis through its centre of mass plus a translational motion of the centre of mass. As it rolls down, it suffers loss in gravitational potential energy provided translational energy due to frictional force is converted into rotational energy.

3. ANS - 2



Let V be the speed of velocity of COM of the Ring just after the impulse is applied and v' is the velocity when pure rolling starts.

Angular velocity ω of the ring at this instant

$$\text{will be } \omega = \frac{v'}{r}$$

Now

Impulse = change in linear momentum

$$\text{We have } J = mv \text{ or } v = \frac{J}{m}$$

Between two positions shown in figure force of friction on the ring acts backwards.

Angular momentum of the Ring about bottommost point will remain conserved

$$\therefore L_i = L_f \text{ or}$$

$$mvr = mv'r + I\omega = mv'r + \frac{(mr^2)v'}{r} = 2mv'r$$

$$\therefore v' = \frac{v}{2} = \frac{J}{2m}$$

4. ANS - 1

$$\text{Given, } \omega_1 = 2\pi \times (400) \text{ rads}^{-1}$$

$$\omega_2 = 2\pi \times (200) \text{ rads}^{-1}$$

$$\alpha = 2\pi \frac{(400-200)}{2} = 200\pi \text{ rads}^{-2}$$

α is the retardation.

5. ANS - 1

Force does not produce any torque because it passes through the center (point of rotation) and we know that if $\tau = 0$ the $L = \text{constant}$.

6. ANS - 4

Total MI of the system,

$$I = I_1 + I_2 + I_3$$

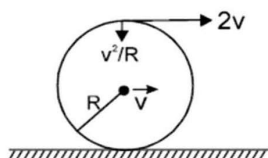
$$I_2 = I_3 = \frac{2}{3}mr^2 + mr^2 = \frac{5}{3}mr^2$$

$$I_1 = \frac{2}{3}mr^2$$

$$I = 2 \times \frac{5}{3}mr^2 + \frac{2}{3}mr^2 = 4mr^2$$

7. ANS - 1

8. ANS - 3



Radius of Curvature is given by

$$= \frac{(\text{velocity})^2}{\text{Normal Acceleration}} = \frac{(2v)^2}{v^2/R} = 4R$$

9. ANS - 1

Moment of inertia of a disc about diameter is:

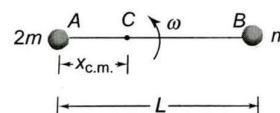
$$\frac{1}{2}MR^2 = I$$

$$\therefore MR^2 = 4I$$

$$\text{Now required moment of inertia} = \frac{3}{2}MR^2 =$$

$$\frac{3}{2}(4I) = 6I$$

10. ANS - 1



Location of centre of mass,

$$X_{cm} = \frac{2m \times 0 + m \times L}{2m + m} = \frac{L}{3}$$

The velocity of ball, A w.r.t. to CM C is -

$$V_A = \omega X_{CM} = \frac{\omega L}{3}$$

$$V_B = \omega(L - X_{CM}) = \frac{2\omega L}{3}$$

Thus, angular momentum of a system about C,

$$L = 2mv_A X_{CM} + mv_B(L - X_{CM})$$

$$= 2m \times \frac{\omega L}{3} \times \frac{L}{3} + m \times \frac{2\omega L}{3} \times \frac{2L}{3} = \frac{2m\omega L^2}{3}$$

11. ANS - 1

The velocity can be computed using

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 2 \\ 3 & 4 & 0 \end{vmatrix} = -8\hat{i} + 6\hat{j} - 3\hat{k}$$

12. ANS - 3

$$\frac{I_A}{I_B} = \frac{\frac{1}{2} m_A r_A^2}{\frac{1}{2} m_A r_B^2} = \frac{m_A r_A^2}{m_A r_B^2}$$

$$\frac{I_A}{I_B} = \frac{\pi r_A^2 t_A r_A^2}{\pi r_B^2 t_B r_B^2}$$

$$\frac{I_A}{I_B} = \frac{t_A r_A^4}{t_B r_B^4} = \left(\frac{4}{1}\right) \left(\frac{1}{4}\right)^4 = \frac{1}{64}$$

$$\therefore I_A < I_B$$

13. ANS - 1

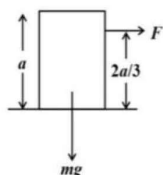
(1) As distance of surface is maximum from axis at equator, hence v is maximum as $\vec{v} = \vec{\omega} \times \vec{r}$

(2) Angular velocity is same everywhere

(3) As distance of surface is maximum from axis at equator, hence v is maximum as $\vec{v} = \vec{\omega} \times \vec{r}$

(4) As distance of surface is maximum from axis at equator, hence v is maximum as $\vec{v} = \vec{\omega} \times \vec{r}$

14. ANS - 2

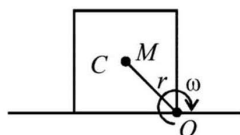


Balancing torque about point o

$$Mg (a/2) = F (2a/3),$$

$$F = 3/4 mg$$

15. ANS - 1



I_0 is moment of inertia of block about an axis through O and perpendicular to the plane of block.

If I_c is moment of inertia about center of mass then

$$I_c = \frac{Ma^2}{6}$$

Hence from parallel axis theorem

$$I_0 = I_c + Mr^2 \text{ and } r^2 = \left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2 = \frac{a^2}{2}$$

$$I_0 = \frac{Ma^2}{6} + \frac{Ma^2}{2} = \frac{2Ma^2}{3}$$

From the conservation of angular momentum about O

$$Mv \frac{a}{2} = \frac{2}{3} Ma^2 \omega$$

$$\omega = \frac{3v}{4a}$$

16. ANS - 3

$$\theta = 2\pi n = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$2\pi \times 10 = \frac{1}{2} \alpha 4^2 \text{ or } \alpha = \frac{40\pi}{16}$$

Let it make N rotation in the first 8 sec

$$\text{Then, } 2\pi N = \frac{1}{2} \alpha 8^2$$

$$N = \frac{1}{2\pi} \times \frac{64}{2} \times \frac{40\pi}{16} = 40$$

The required number of rotations

$$= 40 - 10 = 30$$

17. ANS - 4

$$t = \sqrt{\frac{2S}{g \sin \theta} \left(1 + \frac{K^2}{R^2}\right)}$$

$$\frac{t_1}{t_2} = \sqrt{\frac{1 + \frac{2}{5}}{1 + \frac{1}{2}}} = \sqrt{\frac{14}{15}}$$

18. ANS - 1

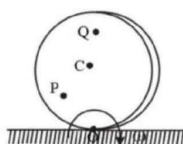
$$\omega = \frac{2 \sin \theta}{5} = \frac{2 \times 4}{5 \times 5} = \frac{8}{25}$$

$$\vec{\omega} = \frac{8}{25} \hat{k} \text{ rad/s}$$

19. ANS - 1

$$\frac{\text{Rotational KE}}{\text{Translational KE}} = \frac{\frac{1}{2} I \omega^2}{\frac{1}{2} m v^2} = \frac{K^2}{R^2}$$

20. ANS - 1



In case of pure rolling bottommost point is the instantaneous center of zero velocity. Velocity of any point on the disc, $V = r\omega$, where r is the distance of the point from O

$$r_Q > r_C > r_P$$

$$V_Q > V_C > V_P$$

21. ANS - 2

Angular momentum about the point of contact is conserved because torque due to friction about point of contact is zero.

22. ANS - 1

The mass of the whole disc = $4M$

Moment of inertia of the disc about the

$$\text{Given axis} = \frac{1}{2} 4MR^2 = 2MR^2$$

Moment of inertia of quarter section of the disc

$$= \frac{1}{4} \times 2MR^2 = \frac{1}{2} MR^2$$

23. ANS - 3

$$\therefore \omega = \sqrt{\frac{\mu g}{r}} \therefore \omega \propto \frac{1}{\sqrt{r}} = r \propto \frac{1}{\omega^2}$$

Since ω is double thus,

$$r' = \frac{4r}{4} = r$$

24. ANS - 4

Solid and hollow balls can be distinguished by any of the three methods.

Here we have, $I_h > I_s$. When torques are equal, angular acceleration a of hollow must be smaller than a of solid. Similarly, on rolling, solid ball will reach the bottom before the hollow ball

25. ANS - 3

$$\omega_1 = 20 \text{ rad s}^{-1}, \omega_2 = 0, t = 4 \text{ s}$$

$$\text{SO angular retardation } \alpha = \frac{\omega_1 - \omega_2}{t}$$

$$= \frac{20}{4} = 5 \text{ rad s}^{-1}$$

Now angular speed after 2 s,

$$\omega_2 = \omega_1 - \alpha t = 20 - 5 \times 2 = 10 \text{ rad s}^{-1}$$

Work done by torque in 2 s

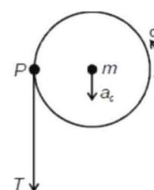
= loss in kinetic energy

$$= \frac{1}{2} I (\omega_1^2 - \omega_2^2) = \frac{1}{2} (0.20) ((20)^2 - (10)^2)$$

$$= \frac{1}{2} (0.20) (400 - 100)$$

$$= \frac{1}{2} (0.20) \times (300) = \frac{1}{2} \times 60 = 30 \text{ J}$$

26. ANS - 3



$$T = ma_c$$

Calculate acceleration of each in ground frame

$$T \cdot R = mR^2 \alpha$$

$$T = mR\alpha$$

$$a_p = (a_c + \alpha R)$$

$$T = ma_c \Rightarrow a_c = \alpha R$$

$$a_p = \frac{2T}{m}$$

17.ANS - 4

Before Collision

$$L_{in} = mv_0 R + \frac{2}{5} mR^2 \omega = \frac{7}{5} mv_0 R$$

It collides elastically with frictionless vertical wall so velocity reverses and angular velocity remains same

After Collision

$$L_{final} = mv_0 R - \frac{2}{5} mR^2 \omega = \frac{3}{5} mv_0 R$$

$$\text{Required Ratio} = \frac{7}{3}$$

18.ANS - 4

$$a_t = \alpha_A r_A = \alpha_C r_C$$

$$\alpha_C = \alpha_A \left(\frac{r_A}{r_C} \right) = 1.6 \times \frac{10}{25} = 0.64 \frac{\text{rad}}{\text{s}^2}$$

$$t = \frac{\omega_C}{\alpha_C} = \frac{\frac{100 \times 2\pi}{60}}{0.64} = 16.35 \text{ s}$$

19.ANS - 4

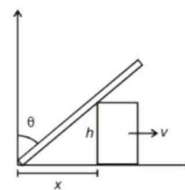
$$TE_i = TE_r$$

$$\frac{1}{2} I \omega^2 = mgh$$

$$\frac{1}{2} \times \frac{1}{3} ml^2 \omega^2 = mgh$$

$$\Rightarrow h = \frac{1}{6} \frac{l^2 \omega^2}{g}$$

30.ANS - 1



$$\tan \theta = \frac{x}{h}$$

$$x = h \tan \theta$$

$$\frac{dx}{dt} = h \sec^2 \theta \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = \frac{v}{h} \cos^2 \theta$$

31.ANS - 3

$$F - f_r = ma \dots\dots\dots (i)$$

$$f_r R = I \alpha = \frac{mR^2}{2} \alpha \dots\dots\dots (ii)$$

For pure rolling

$$a = R\alpha \dots\dots\dots (iii)$$

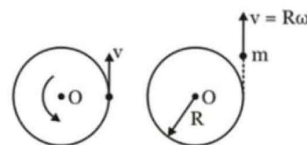
From (i), (ii), and (iii)

$$F - \frac{mR\alpha}{2} = m\alpha R$$

$$F = \frac{3}{2} mR\alpha$$

$$\alpha = \frac{2F}{3mR}$$

32.ANS - 1



Since the broken away piece of mass m has the same linear speed as it had before detaching from the disc, the angular speed of the remaining disc will remain unchanged.

33. ANS - 1

$$mv_0 = MV_{cm} \Rightarrow V_{cm} = mv_0/M$$

$$mv_0x = \frac{1}{12}ML^2\omega \Rightarrow \omega = \frac{12mv_0x}{ML^2}$$

$$V_{AG} = 0 \Rightarrow V_{AC} + V_{CG} = 0$$

$$-\omega \frac{L}{2} + \frac{mv_0}{M} = 0 \Rightarrow \frac{mv_0}{M} = \frac{12mv_0x}{ML^2} \times \frac{L}{2}$$

$$\Rightarrow x = L/6$$

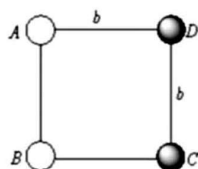
34. ANS - 4

$$KE = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

$$KE = \frac{1}{2}Mv^2 + \frac{1}{2}I\left(\frac{v}{2R}\right)^2$$

$$\text{Where } I = \frac{5}{2}MR^2 \text{ so } KE = \frac{13}{16}Mv^2$$

35. ANS - 3



Moment of inertia of a hollow sphere of radius R about the diameter passing through D is

$$I_A = I_B = \frac{2}{3}MR^2 \dots (i)$$

$$I_C = I_D = \frac{2}{5}MR^2 \dots (ii)$$

Moment of inertia of whole system about side

$$AD = I_A + I_D + I_B + I_C$$

$$= \frac{2}{3}MR^2 + \frac{2}{5}MR^2 + (Mb^2 + \frac{2}{3}MR^2) + (Mb^2 + \frac{2}{5}MR^2)$$

$$= \frac{32}{15}MR^2 + 2Mb^2$$

36. ANS - 4

Mass and distribution of mass about the axis

37. ANS - 2

The rod will rotate about A with

Angular acceleration :

$$\alpha = \frac{\tau}{I} = \frac{F \times x}{\frac{ml^2}{3}} = \frac{3Fx}{ml^2}$$

$$\therefore a_c = \frac{l}{2}\alpha = \frac{3Fx}{2ml}$$

or $a \propto x$ i.e.

$a - x$ graph is a straight line passing through origin.

38. ANS - 2

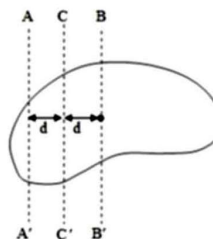
$$F \times 2R = \frac{7}{5}MR^2 \times \left(\frac{a_e}{R}\right)$$

$$\Rightarrow a_e = \frac{10F}{7M}$$

$$\therefore F + f_s = m \times a_e$$

$$\Rightarrow f_s = \frac{3F}{7}$$

39. ANS - 3



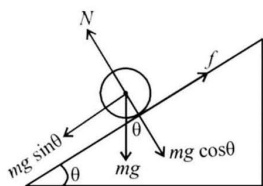
Let BB' be an axis passing through the centre of mass and parallel to CC'

$$\therefore I_{AA'} = I_{BB'} + 4Md^2$$

$$= I_{CC'} - Md^2 + 4Md^2$$

$$I_{AA'} = I_{CC'} + 3Md^2$$

40. ANS – 2



$$mg \sin \theta - f = ma$$

$$fR = \frac{2}{3}mR^2\alpha$$

for pure rolling $a = R\alpha$

Hence, $fR = \frac{2}{3}mR^2\alpha = \frac{2}{3}mRa$

$$\Rightarrow f = \frac{2}{3}ma \Rightarrow f = \frac{2}{5}mg \sin \theta \dots \dots (1)$$

$$f_{\max} = \mu mg \cos \theta$$

$$f \leq f_{\max}$$

$$\Rightarrow \mu_{\min} = \frac{2}{5} \tan \theta$$

TOPIC : GRAVITATION

Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55
A.	3	4	3	3	3	2	1	2	3	2	3	2	4	1	3
Q.	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	2	2	1	2	2	3	1	1	1	2	2	4	1	3	2
Q.	71	72	73	74	75	76	77	78	79	80					
A.	1	1	2	4	2	2	3	3	3	1					

HINT & SOLUTION

41. ANS - 3

$$p = mv_0 = m \sqrt{\frac{GM}{r}} \quad \text{i.e. } p \propto 1/\sqrt{r}$$

$$\frac{p_A}{p_B} = \sqrt{\frac{Rb}{Ra}} = \sqrt{\frac{R+3R}{R+R}} = \sqrt{2}$$

42. ANS - 4

$$g = \frac{GM_p}{R_p^2} = \frac{G}{R_p^2} \cdot \frac{4\pi}{3} R_p^3 \rho$$

$$g = \frac{4}{3} \pi G R_p \rho$$

$$\text{So, } \frac{g_1}{g_2} = \frac{\frac{4}{3} \pi G R_1 \rho_1}{\frac{4}{3} \pi G R_2 \rho_2}$$

$$\Rightarrow \frac{g_1}{g_2} = \frac{R_1 \rho_1}{R_2 \rho_2}$$

43. ANS - 3

As we know, gravitational potential

$$V = -\frac{GM}{R}$$

Gravitational potential at mid-point,

$$= -\frac{G \times 10}{1} - \frac{G \times 100}{1} = -110G$$

$$= -110 \times 6.67 \times 10^{-11} \text{Jkg}^{-1} = -7.3 \times 10^{-9} \text{Jkg}^{-1}$$

44. ANS - 3

When only conservative forces are acting, mechanical energy is conserved and at maximum height speed is zero

$$\frac{-GMm}{R} + \frac{1}{2} m \left(\frac{v_e}{2} \right)^2 = U + 0$$

$$\frac{-GMm}{R} + \frac{1}{2} m \left(\frac{GM}{2R} \right) = U$$

$$U = \frac{-GMm}{R} + \frac{GMm}{4R} = \frac{-3GMm}{4R}$$

45. ANS - 3

According to kepler's third law,

$$\text{Since, } T^2 = kr^3$$

For small changes

$$\frac{2\Delta T}{T} = \frac{3\Delta r}{r}$$

$$= \frac{\Delta T}{T} = \frac{3}{2} \frac{\Delta r}{r}$$

46. **ANS - 2**

$$\text{Total energy} = -\frac{GMm}{2r}$$

$$\text{Here, } r = R + h, GM = g_0 R^2$$

$$E = \frac{-mg_0 R^2}{2(R+h)}$$

47. **ANS - 1**

$$I = \left| \frac{\Delta \vec{V}}{\Delta r} \right| = \frac{8-4}{(1) \sin 30^\circ} = 8 \text{ Nkg}^{-1}$$

48. **ANS - 2**

49. **ANS - 3**

At the surface of the earth, the weight of the person.

$$W = mg = \frac{mGM}{R^2}$$

At a height h

$$\text{Weight } W' = mg' = \frac{mGM}{(R+h)^2}$$

$$\therefore \frac{W'}{W} = \frac{R^2}{(R+h)^2}$$

$$\text{When } h=R, \frac{W'}{W} = \frac{R^2}{(2R)^2} = \frac{1}{4}$$

50. **ANS - 2**

Let us assume that the maximum distance of the body from the centre of the earth is h.

Applying the conservation of mechanical energy, we get

$$\frac{1}{2} m \left(\sqrt{\frac{9GM}{8R}} \right)^2 - \frac{GMm}{R} = -\frac{GMm}{h}$$

$$\Rightarrow -\frac{7}{16} \frac{GMm}{R} = -\frac{GMm}{h}$$

$$\Rightarrow h = \frac{16R}{7}$$

51. **ANS - 3**

The acceleration due to gravity at a distance x ($x < R$) from centre of earth

$$g' = g \frac{x}{R} \Rightarrow g' = g \frac{R/2}{R} = \frac{g}{2}$$

52. **ANS - 2**

$$\text{Because, } v_e = \sqrt{\frac{2GM_e}{R}}$$

The above formula shows that escape velocity is independent of the mass of the projectile.

53. **ANS - 4**

$$\frac{GMm}{R} = 54 \Rightarrow \frac{GM}{R} = \frac{54}{3} = 18$$

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{36} = 6 \text{ m/s}$$

54. **ANS - 1**

At equator

$$g' = g - \omega^2 R$$

when earth suddenly stops $\omega = 0$

$$g' = g - (0)^2 R$$

$g' = g$, This means g will increase.

55. **ANS - 3**

Using Kepler's 3rd law of planetary motion,

$$T = 2\pi \sqrt{\frac{r^3}{GM}} \text{ or } T^2 \propto r^3$$

$$\text{Frequency, } f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{GM}{r^3}}$$

Frequency is independent of the mass of the satellite.

Hence, required ratio = 1:1

56. ANS - 2

Given,

$$F = KR^{-5/2}$$

We know, $F = m\omega^2 R$

$$\omega^2 = R^{-7/2}$$

$$T^2 \propto R^{7/2} \text{ (since } T \propto 1/\omega \text{)}$$

57. ANS - 2

Gravitational acceleration is given by

$$g = \frac{GM}{R^2}$$

Where G = gravitational constant

$$\therefore \frac{g}{G} = \frac{M}{R^2}$$

58. ANS - 1

$$v_1 r_1 = v_2 r_2 = \text{constant}$$

$$\therefore v_0 d = 3dv_1$$

$$v_1 = \frac{v_0}{3}$$

$$\therefore KE = \frac{1}{2} \frac{mv_0^2}{9}$$

$$KE = \frac{k_0}{9}$$

59. ANS - 2

Acceleration due to gravity at a height h from earth's surface

$$g' = \frac{GM}{(R+h)^2}$$

$$\text{since } g' = \frac{g}{100}$$

$$\text{or } \frac{g}{100} = \frac{GM}{(R+h)^2}$$

$$\text{or } \frac{(R+h)^2}{100} = \frac{GM}{g}$$

$$\frac{(R+h)^2}{100} = R^2 \left(\text{since } g = \frac{GM}{R^2} \right)$$

$$R+h = 10R$$

$$h = 9R$$

60. ANS - 2

$$\frac{(v_e)p_1}{(v_e)p_2} = \frac{\sqrt{2g_1 R_1}}{\sqrt{2g_2 R_2}} = \sqrt{\frac{g_1}{g_2}} \times \frac{R_1}{R_2} = \sqrt{ab}$$

61. ANS - 3

The gravitational field inside the cavity is uniform and at the centre of the cavity, it can be calculated as

$$E_{\text{Total}} - E_{\text{cavity}} = \frac{GMR}{R^3} - 0 = \frac{GM}{2R^2}$$

$$F = \frac{GMm}{2R^2}$$

62. ANS - 1

Velocity at the surface = $v_s = \eta v_e$

Velocity in the orbit = v_0

By conservation of mechanical energy,

$$\frac{1}{2}mv_s^2 + \left(-\frac{GMm}{R}\right) = \frac{1}{2}mv_0^2 - \frac{GMm}{r} \dots \dots (1)$$

$$\text{By Newton's law, } \frac{GMm}{r^2} = \frac{mv_0^2}{r} \dots \dots \dots (2)$$

Solving the equation (1) and (2)

$$v_0 = v_e \sqrt{1 - \eta^2}$$

63. ANS - 1

A geostationary satellite has a time period of $T = 24$ hr.

From Kepler's Law,

$$T^2 \propto R^3 \Rightarrow \frac{T_2^2}{T_1^2} = \frac{R_2^3}{R_1^3}$$

$$\Rightarrow \frac{T_2^2}{(24)^2} = \frac{(R + 2.5R)^3}{(R + 6R)^3}$$

$$T = \sqrt{\frac{24 \times 24}{8}} = 6\sqrt{2} \text{ hour}$$

64. **ANS - 1**

Both the satellites will have the same time period as the period is independent of the mass of the satellite.

65. **ANS - 2**

$$3 = \frac{GM_2}{2^2}, \quad 2 = \frac{GM_1}{1^2}$$

$$\therefore \frac{3}{2} = \frac{1}{4} \frac{M_2}{M_1}, \quad \frac{M_1}{M_2} = \frac{1}{6}$$

66. **ANS - 2**

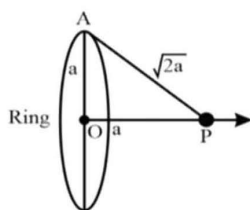
Increase in potential energy is,

$$\Delta U = \frac{mgh}{1 + \frac{h}{R}}$$

$$\therefore h = R$$

$$\text{So, } \Delta U = \frac{mgR}{2}$$

67. **ANS - 4**



The gravitational potential at P

$$V_p = -\frac{GM}{AP}$$

$$V_p = \frac{-GM}{\sqrt{a^2 + a^2}} = \frac{-GM}{\sqrt{2}a}$$

And gravitational potential at O,

$$V_o = -\frac{GM}{a}$$

Let v be the velocity of the particle when it reaches O. K.E. at O = $\frac{1}{2}mv^2$

Gravitational potential energy = Gravitational potential $\times$ mass

$$\text{P.E. at P} = \frac{-GM}{\sqrt{2}a} m \quad \text{and} \quad \text{P.E. at O} = \frac{-GM}{a} m$$

By the principle of conservation of mechanical energy

$$\frac{1}{2}mv^2 - \frac{GMm}{a} = \frac{-GM}{\sqrt{2}a} m$$

$$v^2 = 2 \left[\frac{GM}{a} - \frac{GM}{\sqrt{2}a} \right] = \frac{2GM}{a} \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$v = \sqrt{\frac{2GM}{a} \left(1 - \frac{1}{\sqrt{2}} \right)}$$

68. **ANS - 1**

The acceleration of S is always directed towards the center of the earth

69. **ANS - 3**

Total energy of surface = Total energy at h height

$$\frac{1}{2}m(kv_e)^2 + \left\{ -\frac{GMm}{R} \right\} = 0 + \left\{ -\frac{GMm}{R+h} \right\}$$

$$h = \frac{Rk^2}{1 - k^2}$$

From the center

$$r = R + h = \frac{R}{1 - k^2}$$

70. **ANS - 2**

Work done is same in both case so

$$\Delta U = mgh = mg'h'$$

$$\Rightarrow m \times 9.8 \times 2 = m \times 1.96 \times h'$$

$$\Rightarrow h' = 10 \text{ m}$$

71. ANS - 1

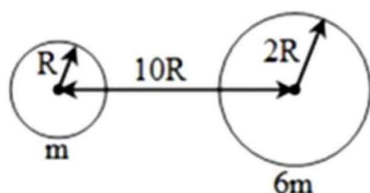
Here, angular momentum is conserved,

i.e., $L = mvr = I\omega$ constant.

$$vr = \text{constant} \Rightarrow v \propto \frac{1}{r}$$

Therefore, the linear speed of planet at A is maximum.

72. ANS - 1



Their relative displacement till collision will be $7R$. Therefore, distance covered by smaller sphere is

$$x_1 = \frac{m_2 d_{rel}}{m_1 + m_2}$$

$$x_1 = \frac{(6m)(7R)}{m + 6m} = 6R$$

73. ANS - 2

Mass of earth $M = 6 \times 10^{24} \text{ kg}$

Mass of super dense material

$$m = 2 \times 10^{24} \text{ kg} = \frac{M}{3}$$

From momentum conservation,

$$MV_1 = \frac{M}{3}V_2 \Rightarrow V_1 = \frac{V_2}{3}$$

From energy conservation,

$$\frac{1}{2}MV_1^2 + \frac{1}{2}\left(\frac{M}{3}\right)V_2^2 = \frac{GM\left(\frac{M}{3}\right)}{\frac{h}{2} + R} - \frac{GM\left(\frac{M}{3}\right)}{h + R}$$

$$v_2 = \sqrt{\frac{3gh}{4}}$$

74. ANS - 4

We know that work done by a conservative force is equal to change in potential energy

$$W = \Delta U = GMm\left(\frac{1}{R} - \frac{1}{3R}\right)$$

$$W = \frac{2GMm}{3R}$$

We know that, $g = \frac{GM}{R^2}$

Hence

$$W = \frac{2mgR}{3}$$

75. ANS - 2

The total energy of the satellite is

$$T.E. = -\frac{GMm}{2r}$$

76. ANS - 2

We know that, ongoing below the earth surface, g decreases, which is given by

$$g' = g\left(1 - \frac{d}{R_e}\right) \dots \dots \dots (1)$$

Where, d = depth

R_e = Radius of earth

At the centre of earth, $d = R_e$ [$\therefore$ from eq. 1]

$$g' = g\left(1 - \frac{R_e}{R_e}\right)$$

$g' = 0$, Hence, weight of the body at the centre,

$$w = mg' = m \times 0 = 0$$

77. ANS - 3

If a satellite is moving in a orbit of $5R$ radius,
The total energy of satellite (binding energy)

$$E_T = -\frac{1}{2} \frac{GMm}{5R}$$

The energy that needs to be provided

= - Binding energy

$$\Rightarrow \text{Energy to be given} = \frac{GMm}{10R}$$

78. **ANS – 3**

From Kepler's Second law aerial velocity remains constant

79. **ANS – 3**

Using Kepler's third law of gravitation, we get

$$\frac{T_n^2}{T_s^2} = \frac{R_n^3}{R_s^3}$$

$$\frac{T_n}{T_s} = \left[\frac{10^{13}}{10^{12}} \right]^{3/2} = (\sqrt{10})^3 = 10\sqrt{10}$$

80. **ANS – 1**

We know that

$$g = \frac{GM}{r^2} = \frac{\frac{G\rho^4}{3} \pi R^3}{R^2} = \frac{4}{3} \pi G \rho R$$

This mean g increases linearly with increase in R, keeping density constant, hence graph is straight line.

TOPIC : FLUID MECHANICS

Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95
A.	2	3	1	1	1	1	4	1	2	1	2	3	2	1	1
Q.	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	3	2	2	4	1	1	2	3	4	4	1	3	1
Q.	111	112	113	114	115	116	117	118	119	120					
A.	3	2	2	4	2	2	2	1	3	2					

HINT & SOLUTION

81. Ans. 2

$$B = \frac{P}{\frac{\Delta V}{V}} = \frac{1}{K}$$

$$\frac{\Delta Y}{V} = KP$$

82. Ans. 3

$$f_1 = k(x_1 - x)$$

$$f_2 = k(x_2 - x)$$

$$\frac{f_1}{f_2} = \frac{x_1 - x}{x_2 - x}$$

$$f_1 \ell_2 - f_1 x = f_2 \ell_1 - f_2 x$$

$$\left(\frac{f_1 \ell_2 - f_2 \ell_1}{f_2 - f_1} \right)$$

83. Ans. 1

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{\frac{20}{10^{-6}}}{\frac{10^{-4}}{1}} = \frac{20}{10^{-6}} \times \frac{1}{10^{-4}}$$

84. Ans. 1

$$f = \left(\frac{A\gamma}{\ell} \right) x$$

$$\left(\frac{(\text{dia})^2}{\ell} \right)$$

85. Ans. 1

$$f_1 = K(\ell_1 - \ell_0) = 4$$

$$f_2 = K(\ell_2 - \ell_0) = 6$$

$$\frac{\ell_1 - \ell_0}{\ell_2 - \ell_0} = \frac{2}{3}$$

$$3\ell_1 - 3\ell_0 = 2\ell_2 = 2\ell_0$$

$$\ell_0 = 3\ell_1 - 2\ell_2$$

$$q = K(\ell_3 - \ell_0)$$

$$q = \frac{4(\ell_3 - \ell_0)}{\ell_1 - \ell_0}$$

$$q\ell_1 - q\ell_0 = 4\ell_3 - 4\ell_0$$

$$4\ell_3 = q\ell_1 - 5\ell_0$$

$$= q\ell_1 - 5(3\ell_1 - 2\ell_2)$$

$$4\ell_3 = 10\ell_2 - 6\ell_1$$

$$\ell_3 = 2.5\ell_2 - 1.5\ell_1$$

86. Ans. 1

$$\rho V = \text{const.}$$

$$\rho' V' = \rho V$$

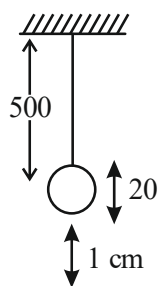
$$\frac{\rho'}{\rho} = \frac{V}{V'}$$

$$\frac{\rho'}{\rho} = \frac{V}{V + \Delta V} = \frac{V}{V \left(1 + \frac{\Delta V}{V} \right)} \quad \dots (1)$$

$$\frac{\Delta V}{V} = \frac{-P}{B} \quad \dots (2)$$

$$\frac{\rho'}{\rho} = \frac{1}{1 - \frac{P}{B}}$$

87. Ans. 4



$$T = mg + \frac{mv^2}{R}$$

$$T = (25 \times 10) + \frac{25 \times v^2}{5.1}$$

$$\Delta \ell = \frac{f \ell}{AY}$$

$$(10^{-2}) = \frac{(250 + 4.9v^2)}{AY} \times 5$$

$$(10^{-2}) = \frac{(250 + 4.9v^2) \times 5}{\pi (0.05)^2 \times 10^{-4} \times 2 \times 10^{11}}$$

$$(250 + 4.9v^2) = \frac{\pi \times 0.05 \times 0.05 \times 10^5 \times 2}{5}$$

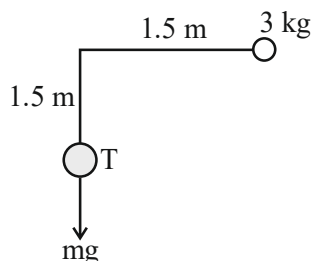
$$250 + 4.9v^2 = 50\pi \times 2$$

$$4.9v^2 = 314 - 250$$

$$v^2 = 13.06$$

$$v = 3.7 \text{ m/s}$$

88. Ans. 1



$$T - mg = mw^2R$$

$$T = mg + mw^2R$$

$$T = 30 + 3 \times (2\pi \times 2)^2 (1.5)$$

$$Y = \frac{T/A}{\Delta \ell / \ell}$$

$$\Delta \ell = \frac{f \ell}{AY}$$

$$\Delta \ell = \frac{740 \times 1.5}{\pi (1 \times 10^{-3})^2 \times 2 \times 10^{11}}$$

$$\Delta \ell = 176 \times 10^{-5} = 1.76 \times 10^{-3}$$

89. Ans. 2

$$25 = T_1 = mg - f_B \quad \dots (1)$$

$$75 = T_2 = f_B + m_\ell g + m_B g \quad \dots (2)$$

$$100 = m_A g + (m_\ell + m_B)g$$

$$100 = m_A g + (25)$$

$$m_A g = 75 \rightarrow \text{Read of D}$$

90. Ans. 1

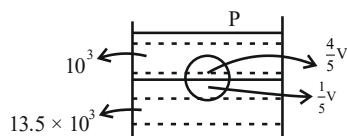
$$\rho_{\text{oil}} gh_1 = \rho_w gh_2$$

$$\frac{\rho_{\text{oil}}}{\rho_w} = \frac{h_2}{h_1}$$

91. Ans. 2

wrt surface speed of top & bottom is same. So fall at the rate of 2 cm/hr.

92. Ans. 3



$$mg = f_{B1} + f_{B2}$$

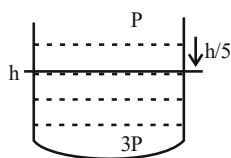
$$\rho v g = \left(10^3 \times \frac{4}{5} v \times g \right) + \left(13.5 \times \frac{1}{5} v \times g \right) 10^3$$

$$= v g (10^3) \left[\frac{4}{5} + \frac{13.5}{5} \right]$$

$$\rho v g = v g (10^3) \left(\frac{17.5}{5} \right)$$

$$\rho = 3.5 \times 10^3$$

93. Ans. 2



$$3P = P + \rho g h$$

$$P' = P + \rho g \left(\frac{4h}{5} \right)$$

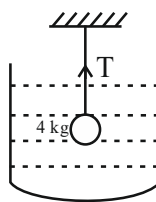
$$P' = P + 2P \left(\frac{4}{5} \right)$$

$$P' = \frac{13P}{5}$$

94. Ans. 1

No unthrust (No pressure difference)

95. Ans. 1



$$mg = 40$$

$$(10000 \times v) = 4$$

$$T + f_B = mg$$

$$T = mg - f_B = 40 - (4 \times 10^{-4} \times 720 \times 10) = 40 - 2.88$$

$$T = 37.12 \text{ N}$$

96. Ans. 1

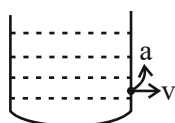
$$mg = f_{B1} + f_{B2}$$

$$(8560 \times 10^3 \times 10^{-6} \times g) = (13600x \times 10^2 \times 10^{-6})g + [1000(10-x)10^2 \times 10^{-6}]g$$

$$8.56 = 13.6x + 10 - x$$

$$x = 6$$

97. Ans. 3



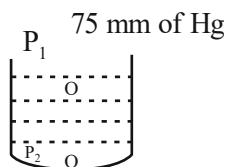
$$f_t = v_r \frac{dm}{dt} = v_r \left[\frac{d}{dt} \rho (a \times x) \right]$$

$$f_t = \rho a v_r^2$$

$$u(mg) = \rho a (2gh)$$

$$u = \frac{2a}{A}$$

98. Ans. 3



$$P_1 V_1 = P_2 V_2 \text{ (Isothermal process)}$$

$$\left[\rho g (75) (3V) \right] = \left[\rho g \left(75 + \frac{h}{10} \right) \right] (V)$$

$$225 = 75 + \frac{h}{10}$$

$$1500 = h = h = 15\text{m}$$

99. Ans. 2

$$\rho_s (AL)g = \left(2\rho A \frac{L}{4} \right)g + \rho \left(A \frac{3L}{4} \right)g$$

$$\rho_s = \frac{\rho}{2} + \frac{3\rho}{4}$$

$$\rho_s = \frac{5\rho}{4}$$

100. Ans. 2

$mg = \text{wt of liquid displaced}$

$$m(10) = (10^3 \times 3 \times 10^2 \times 10^{-6}) \times 10$$

$$= 0.3 \text{ kg} = 300 \text{ gm}$$

101. Ans. 4

More than atm pressure.

102. Ans. 1

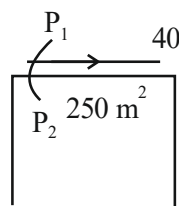
$$V \propto r^2 \quad 2\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$V' \propto R^2$$

$$\frac{V'}{V} = \frac{R^2}{r^2} = V' = V \frac{(2r^3)^{2/3}}{r^2} = 5(4)^{1/3}$$

103. Ans. 1

104. Ans. 2



$$P_2 + \rho gh_2 + \frac{1}{2}\rho V_2^2 = P_1 + \rho gh_1 + \frac{1}{2}\rho V_1^2$$

$$P_2 - P_1 = \frac{1}{2}(1.2) \times 40 \times 40$$

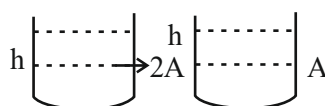
$$f = (P_2 - P_1)A = \frac{1}{2} \times 1.5 \times 40 \times 40 \times 250$$

$$f = 2.4 \times 5 \times 10^5$$

105. Ans. 3

At mid point of surface and level group $y = h$

106. Ans. 4



$$\rho_1 (2A \times \sqrt{2gh}) = \rho_2 (A \times \sqrt{2gh})$$

$$\frac{\rho_1}{\rho_2} = \frac{1}{2}$$

107. Ans. 4

$$6\pi\eta r v = \frac{4}{3}\pi r^3 (\rho_0 - \rho_\ell)g$$

$$V = \frac{2}{9} \frac{r^2 g}{x} (\rho_0 - \rho_\ell)$$

$$\frac{r_1^2 g}{\eta} (8 \times 10^3 - 2 \times 10^3) = \frac{r_2^2 g}{\eta} (11 \times 10^3 - 2 \times 10^3)$$

$$\frac{r_1^2}{r_2^2} = \frac{3a \times 10^3}{26 \times 10^3} = \frac{r_1}{r_2} = \sqrt{\frac{3}{2}}$$

108. Ans. 1

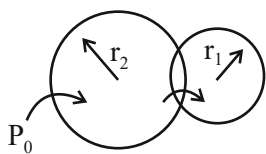
$$P_1 + \rho gh_1 + \frac{1}{2} \rho V_1^2 = P_2 + \rho gh_2 + \frac{1}{2} \rho V_2^2$$

$$P + \frac{1}{2} \rho V^2 = \frac{P}{2} + \frac{1}{2} \rho V_2^2$$

$$\frac{P}{2} / \frac{\rho}{2} + V_1^2 = V_2^2$$

$$\sqrt{\frac{P}{\rho} + v^2} = v_2$$

109. Ans. 3



$$P_0 + \frac{4T}{r_2} + \frac{4T}{r} = P_0 + \frac{4T}{r_1}$$

$$\frac{1}{r} = \frac{1}{r_1} - \frac{1}{r_2}$$

$$r = \frac{r_1 r_2}{r_2 - r_1}$$

$$r_1 = r_2 \quad r = \infty$$

110. Ans. 1

$$T(4\pi R)$$

$$(73 \times 10^{-7}) \left(4\pi \times \frac{1.5}{100} \right)$$

$$= 1.37 \times 10^{-2} \text{ N}$$

111. Ans. 3

$$\frac{4}{3} \pi R^3 = 125 \frac{4}{3} \pi r^3$$

$$R = 5r$$

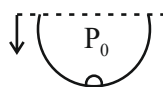
$$\Delta U = T(125 \cdot 4\pi r^2 - 4\pi R^2)$$

$$= T(125(4\pi)r^2 - 4\pi(5r)^2)$$

$$= T \cdot 4\pi(100r^2)$$

$$\Delta U = 16 \pi R^2 T$$

112. Ans. 2



$$P_0 + \frac{2T}{r} = P_0 + \rho gh$$

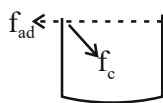
$$\frac{2T}{r} = \rho gh$$

$$r = \frac{2T}{\rho gh} = \frac{2 \times 0.07 \times 100}{1000 \times 10 \times 40}$$

$$r = \frac{7}{2} \times 10^{-5}$$

$$a = 7 \times 10^{-5} = 0.07 \text{ mm}$$

113. Ans. 2



$$\frac{F_c}{F_2} = F_A$$

114. Ans. 4

$$n \frac{u}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow n = \frac{R^3}{r^3}$$

$$KL = nT 4\pi r^2 - 4\pi R^2 T$$

$$= 4\pi T (nr^2 - R^2)$$

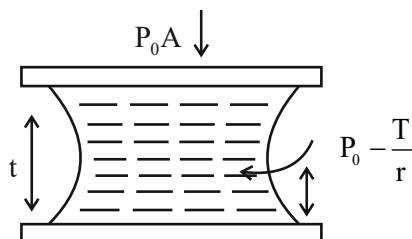
$$= 4\pi T \left(\frac{R^3}{r^3} (r^2) - R^2 \right)$$

$$= 4\pi T \left(\frac{R^3}{r} - R^2 \right)$$

$$\frac{1}{2} \left[\frac{4}{3} \pi R^3 \rho \right] v^2 = 4\pi T R^3 \left(\frac{1}{r} - \frac{1}{R} \right)$$

$$v = \sqrt{\frac{6T}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$$

115. Ans. 2



$$F_{\text{excess}} = \left(\frac{T}{r} \right) \text{Area} \times \frac{2A}{2A}$$

$$= \frac{T}{V_0} \times 2A^2$$

$$F = \frac{(70) \times 2 \times 40 \times 40 \times 100}{0.05}$$

$$\frac{56 \times 4 \times 10^5}{5} \text{ dyne}$$

$$F = 49.8 \text{ N}$$

116. Ans. 2

$$1 \times 1 \times 1 = \frac{4}{3} \pi R^3$$

$$R = \left(\frac{3}{4\pi} \right)^{1/3}$$

$$A = 4\pi R^2 = 4\pi \left(\frac{3}{4\pi} \right)^{2/3}$$

$$= \left(\frac{9}{16\pi^2} \times 64\pi^3 \right)^{1/3}$$

$$A = (36\pi)^{1/3}$$

117. Ans. 2

$$T = 7.2 \times 10^{-2} \text{ N / m}$$

$$w = T(2A)$$

$$= T(2 \times 10 \times 1 \times 10^{-5})$$

$$= (7.2 \times 10^{-2})(20 \times 10^{-5})$$

$$= 14.4 \times 10^{-5} = 1.44 \times 10^{-5} \text{ J}$$

118. Ans. 1

$$KV^2 + \rho_2 Vg = \rho_1 Vg$$

$$V = \sqrt{\frac{(\rho_1 - \rho_2) Vg}{K}}$$

119. Ans. 3

$$V \propto r^2$$

$$V^1 = 4V$$

120. Ans. 2

$$= xA \frac{\Delta V}{\Delta x}$$

$$= \frac{0.01 \times 100 \times 10}{10^{-1} \times 100} = 100 \text{ dyne}$$